Author: From the viewpoint of the bounded sequence definition, my numbers A and B are not better and not worse than yours. However, your last sentence is peculiar. What do you mean by saying "more accurate"?

Reader: My A is apparently the greatest of all possible lower bounds, while my B is the least of all possible upper bounds.

Author: The first part of your statement is doubtlessly correct, while the second part of it, concerning B, is not so self-explanatory. It needs proof.

Reader: But it seemed rather obvious. Because all the terms of (5) increase gradually, and evidently tend to unity, always remaining less than unity.

Author: Well, it is right. But it is not yet evident that $B=1$ is the least number for which $y_n \leq B$ is valid for all n. I stress the point again: your statement is not self-evident, it needs a proof.

I shall also note that "self-evidence" of your statement about $B=1$ is nothing but your subjective impression; it is not a mathematically substantiated corollary.

Reader: But how to prove that $B=1$ is, in this particular case, the least of all possible upper bounds?

Author: Yes, it can be proved. But let us not move too fast and by all means beware of excessive reliance on so-called self-evident impressions.

- The warning becomes even more important in the light of the fact that the boundedness of a sequence does not imply at all that the greatest A or the least B must be known explicitly. → Ostadhar on fact

- Now, let us get back to our sequences and find other examples of bounded sequences.

Sam: Seq 7 is Bounded. $A=0$, $B=1$. Seq 9 is Bounded. $A=-1$, $B=1$. Seq 10 is Bounded. $A=0$, $B=1$. Seq 13 is Bounded between